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Measuring The Thermal Conductivity of Fabric Materials And Dressings Using A Heat-Flow Meter Under different Humidity conditions

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1 Introduction

The introduction provides the background of pressure ulcers and heat-related illnesses. Also, the introduction helps us understand the impact of different wound dressings and fabric materials when dealing with pressure ulcers and heat strokes, accordingly.

1.1 Pressure Ulcers

In the medical field, deformation damage, also known as a pressure ulcer refers to a type of injury that occurs in the skin or soft underlying tissues. Those injuries tend to appear when a constant, prolonged pressure is exerted on specific body areas for a certain period. [1,2]

Pressure ulcers which are also commonly termed bed-sores, decubitus ulcers, or pressure sores, are caused by applying an external force on a specific area, which could either be shear, compression or both of them combined. This injury type could end up causing restriction or reduction in blood flow to certain tissues (ischemia). Moreover, the supply of both oxygen and nutrients to those tissues experiencing pressure ulcers might be harmed and eventually, this condition could lead to tissue death, also called necrosis. Patients having pressure ulcers should be treated properly since otherwise, their injuries might lead to fatal complications. [2,3]

Once pressure ulcers appear, they tend to heal slowly and they might recur. Thus, preventing them from occurring in the first place has significant importance. There are some materials or equipment that could be used to prevent them from occurring, such as pressure-relieving mattresses and supportive surfaces could lower the risk since they can provide a soft surface that comes in touch with skin and soft underlying tissues and skin and distribute the pressure over a larger surface area. This can lead to decreased pressure which is exerted on body areas, especially vulnerable ones. [4] Moreover, various dressing types could be utilized for many stages of wound healing, since a whole range of different dressings is available to help assist with the different stages of the healing process. The different dressings which are available nowadays could be classified as non-absorbent, absorbent, debriding, self-adhering, and many more. Determining the most appropriate dressing mostly depends on the location or the ulcer type, personal preference, or cost to the patient. Dressings are usually occlusive to enable better healing of the ulcers since they tend to heal better when the environment is moist. When the ulcer is clean and dry, occlusive dressings should be changed weekly. However, if more frequent dressings changes are made, except for removing debris, healthy cells might be removed as well, and thus it should be avoided. [2] Some Dressings can even help reduce the bacterial load and prevent further trauma. At present, there are many new types of dressings including wet dressings, silver ion dressings, foam dressings, hydrocolloid dressings, and others. [5]

1.2 Heat-Related illnesses

Many mechanisms of the human body are associated with the regulation of temperature, including conduction (direct heat transfer between 2 surfaces), convection (heat transfer from solid/liquid to moving liquid or air), radiation (heat transfer by electromagnetic waves), and evaporation (heat dissipation by evaporation of liquid). All of those are designated to maintain core body temperature at 37°C in healthy people. There is a critical thermal maximal temperature range for humans that averages between 41.6°C to 42°C. [6]

Heat-related illnesses refer to several syndromes which are caused by a disruption in the mechanism of temperature in people who were exposed to high environmental heat. [7] If one is exposed to heat in large amounts which exceed the compensatory mechanisms of thermoregulation, it will cause heat-related illnesses [6] Heat-related illnesses have various symptoms, from mild to more severe ones, including heat edema and muscle cramps or collapses (syncopes), both related to exercise, heat exhaustion, and life-threatening heat strokes, with or without multi-organ failure. Some functionaries, such as Athletes, outdoor laborers, and military personnel are more likely to experience heat-related illnesses, resulting in such symptoms. Several factors, both intrinsic and extrinsic ones, might increase the risk to experience heat-related illness, including medical conditions, environmental factors, medication use, and inadequate acclimatization. If the recognition and treatment of the heat-related condition are done properly and effectively, they will help prevent fatal outcomes. Otherwise, if the inflammation is uncontrolled, chronic inflammation might develop. [6,8]

Clothes create a layer of isolation, a barrier that lessens the evaporative heat loss ability by compromising the heat dissipation from the surface of the skin. This barrier could eventually lead to a decreased cooling efficiency and an increase in both skin and core body temperature, mostly during exercise. Certain clothing fabrics could decrease the resistance to evaporation due to properties of high vapor permeability and low thermal insulation. Those could also potentially optimize the balance of temperature regulation and help maintain the rate of heat storage at low levels, around 0, especially while exercising. Moreover, some clothing could even lead to improved exercise performance. [9]

1.3 Thermal Conductivity

An important thermal property of any fabric, also related to its breathability, is called thermal conductivity. This property describes how well heat could flow through a specific substance. A fabric that is characterized by a high thermal conductivity allows for a heat transfer from a hot side, such as the human body to a cooler side such as the air on the other side of clothing. Therefore, If the thermal conductivity is higher, a substance will better conduct heat. [10] In general, the thermal conductivities of textile structures range between 0.033 to 0.10 $\frac{W}{mK}$. [11]

1.4 Thermal Conductivity Measuring Techniques

Nowadays, there are many used techniques designated to measure the thermal conductivity constant of different materials. In general, the various techniques can be divided into two categories, both steady-state and transient methods. One is used to determine thermal conductivity directly, requiring a relatively large specimen and long measuring times. The other uses small specimens, and the measuring time is a few minutes, but only allows for the thermal diffusivity parameter to be measured. [12] For example, one of the techniques is the Constant Temperature Method, based on a simple concept in which a fabric sample is wrapped around a hot body. This method measures the energy amount which is required to maintain the constant temperature gradient between the fabric and coolant (mostly air). The apparatus used for this method is composed of a cylinder heater on which the tested fabric sample should be wrapped while the cylinder ends are properly insulated. The cylinder is maintained at steady-state conditions while it is electrically heated and so the thermal conductivity of the fabric could be calculated in the manner mentioned above. [13]

Another technique for measuring the thermal conductivity constant is by using a thermal camera and image-processing-based tools. Under the assumption of a uniaxial heat flow and steady-state thermal conditions, the thermal conductivity λ can be calculated using this IRT-based method, according to the first Fourier's law of heat conduction. [13]

To date, the thermal conductivity constant has a significant impact in many fields, especially in the wound-healing process and heat-related illnesses. Therefore, measuring this property using statistical means is very useful since the thermal conductivity of the different materials could be very applicable to a variety of uses. This study aims to measure the thermal conductivity constant of different fabric materials and wound dressings, under humidity conditions from wishful thinking that innovative dressings and fabrics with better properties could be planned, analyzed, manufactured, and eventually used in the future to help treat pressure ulcers and prevent heat-related illnesses.

2 Materials and Methods

2.1 Wound dressing

Mepilex Lite is a thin foam dressing that absorbs exudate and maintains a moist wound environment with a silicone adhesion layer for less-pain contact to skin or wound surface. The silicone layer seals around the wound edges, preventing the exudate to leak onto the surrounding skin, thus minimizing the risk for maceration, as well as ensuring atraumatic dressing changes. Mepilex Lite can be cut to suit various wound shapes and locations, and designed for the management of a wide range of non to low-exuding wounds such as leg and foot ulcers, pressure ulcers, partial thickness burns, radiation

skin reactions, and Epidermolysis Bullosa. This dressing can also be used as a protection for compromised and fragile skin. [14]

2.1.1 Hydrogel-based dressings

Hydrogel-based dressings were found to be effective in preventing device-related injuries. The dressings exhibited long-term stiffness and thermal conductivity that matched the corresponding properties of human skin under both dry and moist conditions. Additionally, the dressings demonstrated tacky behavior that minimized frictional skin-device sliding, and skin-matching thermal conductivity, allowing for the transfer of heat potentially accumulated in the skin under the applied device to the environment. These mechanical and thermal properties make hydrogel-based dressings advantageous for the prevention of medical device-related pressure ulcers (MDRPUs) from a bioengineering perspective . [15] The thermal conductivity of hydrogels is within a range of 0.27-0.31 W/mK. [16]

2.1.2 Resposorb Silicone Border (RSB) dressing

Resposorb Silicone Border (RSB) dressing was found to provide the best balance between the protective performance of a new product and its ongoing protective efficacy in comparison to silicone-foam dressings, as it absorbs moisture over time, as measured by the prophylactic trade-off design parameter (PTODP). Although the results demonstrated that the dressing with the fluff core is at least as good as silicone foams, if replaced frequently to maintain dryness, the RSB dressing demonstrated superior biomechanical protective efficacy (PEI) over silicone-foam dressings. [17] The thermal conductivity of foams is within a range of 0.05-0.11 W/mK. [16] Since the dressing that was used for the experiment, Mepilex Lite is made of both foam and hydrogel its thermal conductivity should range within approximately (according to average calculation) 0.16-0.21 W/mK

2.2 Fabric Material

IDF uniform fabric material is 100% cotton. [18] Cotton fabric is a widely used natural textile composed of fibers that are polymers of glucose residues linked by glycosidic bonds. This fabric is characterized by its good air permeability and comfortable softness, making it a staple in everyday life. However, the presence of hydrophilic groups such as hydroxyl groups on the surface of cotton fabric allows it to absorb moisture. As a result, when worn, cotton fabric can readily absorb moisture and provide a conducive environment for bacterial growth. [19] The thermal conductivity of cotton, which represents the thermal conductivity of IDF uniform, ranges between 0.026-0.065 W/mK. [20]

2.3 Heat-Flow meter Apparatus

The thermal conductivity of each tested specimen (fabric/ wound dressing) was calculated using the Heat-Flow meter, based on Fourier's law for a one-dimensional (1D) heat transfer process. The Heat-Flow meter consists of two metal blocks,

measuring 4x4x5cm, with one block composed of aluminum and the other of steel (stainless, type 304).

These two metal blocks were positioned so that the specimen was between them, with the aluminum placed above and the steel placed below. A 1-cm thick, porous rubber sheet was used to insulate the two blocks, which minimized the radial heat flow and imposed a mainly unidirectional heat flux through the device, running perpendicular to the specimen, resulting in a heat flux that was transferred primarily along its thickness axis. Moreover, four thermocouples were embedded in the metal blocks, two on each one, at symmetrically distanced locations. The thermocouples allowed us to acquire the temperature gradient across the blocks and specimen structure. The temperature readings from each thermocouple were digitally and automatically recorded into a computer using the LabVIEW data acquisition software.

The heat-flow meter utilized two thermoelectric coolers that were powered by a 300W AC-DC multiple-switching power supply to heat the lower block and cool the upper block. To simulate sustained loading from and ensure full contact between the metal blocks and the specimen, a weight of 2.5 kg was applied to the top surface of the heat-flow meter, which transferred pressure onto the tested specimen. Figure 1 presents the Heat-Flow meter structure.

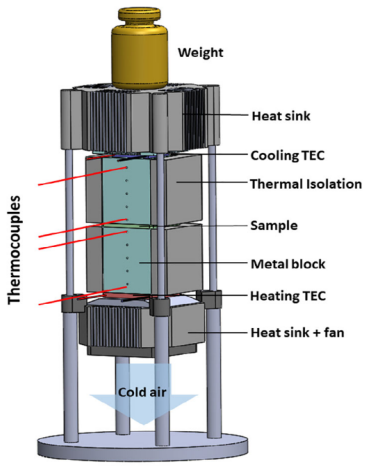

Figure 1- Schematic diagram of the heat flow meter

The system was deemed to have reached a quasi-steady-state thermodynamic condition when the temperature readings from all thermocouples had fluctuated by less than 1% within a 10-minute timeframe. As stated above, the thermal conductivity coefficient k of the tested material (fabric/ wound dressing) was calculated using Fourier's law for a one-dimensional (1D) heat transfer equation, under the assumption that the heat flux along the metal blocks and the tested material is equal:

$$(1)\quad Q = k_{steel} \cdot A \cdot \frac{T_1 - T_2}{X_{steel}} = k_{specimen} \cdot A \cdot \frac{T_2 - T_3}{X_{specimen}} = k_{aluminum} \cdot A \cdot \frac{T_3 - T_4}{X_{aluminum}}$$

Where k_i is the thermal conductivity of each material, X_i is the thickness of each material, A is the area of the tested specimen in contact with the superior/inferior blocks, T_1 is the temperature of the lowest thermocouple placed in the inferior steel

block, T_2 is the temperature at the inferior surfaces of the tested specimen, T_3 is the temperature at the superior surfaces of the tested specimen and T_4 is of the highest thermocouple placed in the superior aluminum block.

The thermal conductivities of the steel and aluminum blocks [21]:

Table 1- Thermal conductivities

Material	Thermal conductivities (k)
Steel (stainless, type 304)	$14.4\ \frac{W}{mK}$
Aluminum	$240\ \frac{W}{mK}$

In this study, the aim is to find the thermal conductivity of the tested materials, and isolate the parameter of interest yields:

$$(2)\ k_{specimen} = \frac{k_{steel}}{X_{steel}} \cdot X_{specimen} \cdot \frac{T_1 - T_2}{T_2 - T_3}$$

The experiment was performed at room temperature, in two different humidities, dry specimen, and moist specimen, with 10 repeats for each specimen and each humidity (a total of 40 measurements), resulting in 4 thermal conductivities.

2.4 Acquiring different moist environments

As said above, in this experiment two different humidities were tested, dry and moist specimens. To produce a moist sample, a solution of isotonic 0.9% saline was created by dissolving 9gr of NaCl in 1 liter of distilled water, which was boiled for approximately 15 minutes to ensure complete solubility of the solute. After cooling down the saline water to room temperature, and considering that the maximum sweat rate for a healthy adult is 1 mg/min per cm^2 [13], 18 mg of the saline solution was uniformly sprayed on a flat dry cleaning towel with an area of 85.56 cm^2 that was supposed to mimic the human epidermis. Then, the fabric specimen with an area of 16 cm^2 was placed on top of the moist cloth and remained on it so that the entire set-up was hermetically sealed to reduce evaporation. Although, as it appeared during the experiment, the towel couldn't successfully absorb the liquid and was absolutely flooded and thus the specimen was entirely wet and not moist as supposed to. Due to that, the fabric specimen was squeezed as much as possible until it was moist and not entirely wet. Afterward, to measure the amount of saline solution that was absorbed into the specimen, the weight of an identically dry specimen was measured as well as the weight of the moist specimen and the difference between the weights measured was considered to be the weight of the saline solution. The dry specimen weight was 1.93 g and the moist specimen weight was 3.6 g so the saline solution weight was 1.67 g. As 1 g saline is equal to 1 ml saline, 1.67 ml of saline solution remained on the specimen.

Following the method conducted in the predicate experiment [13], which used the maximum sweat rate for healthy adults while waiting one hour, the following calculation was performed to obtain the waiting time for each sample in our experiment:

$$(3)\ \frac{12ml}{6.42\ cm^2 \cdot 60\ min} = \frac{1.67ml}{A_{specimen} \cdot T}$$

Where $A_{specimen}$ is equal to the area of the specimen, 1.67ml is the actual amount of saline solution on the sample and T is the required waiting time.

Following that equation, and the area of the fabric was 16 cm^2, 3.35 minutes were waited before measuring the fabric specimen. According to the same method, 1.67 ml of saline solution was sprayed on 25 cm^2 of dressing specimen. Using equation 3, 2.14 minutes were waited before measuring the fabric specimen.

2.5 Results and statistical analysis

Temperature measurements were obtained from each repeat, and the thermal conductivity was calculated using Equation 2. Then, the average of each 10 thermal conductivities obtained for each sample was calculated together with the standard deviation. The averaged thermal conductivity for each dry specimen was then compared to its value that appear in the literature. Afterward, statistical analysis was performed with MATLAB software, to compare the thermal conductivity of the dry specimen to the thermal conductivity of the moist specimen, for both materials. Two-tailed t-tests were performed, with a statistical significance of $P < 0.05$, to determine whether there is a difference between the thermal conductivities of the dry and moist specimens.

3 Results

The following tables show the temperatures measured for each specimen by the Heat-Flow Meter and the thermal conductivity calculated according to Equation 2:

*Table SEQ Table * ARABIC 2- Results of the dry dressing*

Dry Dressing									
T1 [°C]	T3 [°C]	T2 [°C]	Room Temp [°C]	Dressing Thickness [cm]	A [cm^2]	Thermal Conductivity-Aluminum [w/mk]	Thermal Conductivity-Steel [w/mk]	Steel Thickness [cm]	Thermal Conductivity-Dressing [w/mk]
60	26	42	26	0.058	16	240	14.4	5	0.1879
60.9	26	42	26	0.058	16	240	14.4	5	0.1973
61	27	42.8	26	0.058	16	240	14.4	5	0.1924
61	27	42.9	26	0.058	16	240	14.4	5	0.1890
61	26	42	26	0.058	16	240	14.4	5	0.1996
60.8	26	42	26	0.058	16	240	14.4	5	0.1963
60.7	26	42	26	0.058	16	240	14.4	5	0.1952
61	26	42	26	0.058	16	240	14.4	5	0.1984
61	27	43	26	0.058	16	240	14.4	5	0.1879
60	26	42	26	0.058	16	240	14.4	5	0.1879
								Average	0.1932
								std	0.0045

Table 3- Results of the moist dressing

Moist Dressing									
T1 [°C]	T3 [°C]	T2 [°C]	Room Temp [°C]	Dressing Thickness [cm]	A [cm^2]	Thermal Conductivity-Aluminum [w/mk]	Thermal Conductivity-Steel [w/mk]	Steel Thickness [cm]	Thermal Conductivity-Dressing [w/mk]
61	39	43	27	0.058	16	240	14.4	5	0.7517
61	39	43	27	0.058	16	240	14.4	5	0.7517
61	38	43	28	0.058	16	240	14.4	5	0.6013
61.8	38	44	26	0.058	16	240	14.4	5	0.4956
62	38	44	27	0.058	16	240	14.4	5	0.5011
62	38	44	26	0.058	16	240	14.4	5	0.5011
62	38	44	26	0.058	16	240	14.4	5	0.5011
62	38	44	26	0.058	16	240	14.4	5	0.5011
62	38	44	26	0.058	16	240	14.4	5	0.5011
62	38	43.8	27	0.058	16	240	14.4	5	0.5242
								average	0.5630
								std	0.1043

Table 4- Results of the Dry Fabric

Dry Fabric									
T1 [°C]	T3 [°C]	T2 [°C]	Room Temp [°C]	Fabric Thickness [cm]	A [cm^2]	Thermal Conductivity-Aluminum [w/mk]	Thermal Conductivity-Steel [w/mk]	Steel Thickness [cm]	Thermal Conductivity-Dressing [w/mk]
63	31	54.8	25	0.037	16	240	14.4	5	0.0367
63.4	31.6	52	25	0.037	16	240	14.4	5	0.0595
66	32	53	24.9	0.037	16	240	14.4	5	0.0660
63	32	52	24	0.037	16	240	14.4	5	0.0586
64.2	32	54	24	0.037	16	240	14.4	5	0.0494
63.7	32	51	24	0.037	16	240	14.4	5	0.0712
64	32	54	24.8	0.037	16	240	14.4	5	0.0484
63.5	32	50.3	25	0.037	16	240	14.4	5	0.0769
63.3	32.1	50	24.2	0.037	16	240	14.4	5	0.0792
65.7	33	50	25	0.037	16	240	14.4	5	0.0984
								Average	0.0644
								std	0.0170

Table 5- Results of the moist fabric

Moist Fabric									
T1 [°C]	T3 [°C]	T2 [°C]	Room Temp [°C]	Fabric Thickness [cm]	A [cm^2]	Thermal Conductivity-Aluminum [w/mk]	Thermal Conductivity-Steel [w/mk]	Steel Thickness [cm]	Thermal Conductivity-Dressing [w/mk]
76.9	25	40	26	0.037	16	240	14.4	5	0.2621
77	25.7	40	27	0.037	16	240	14.4	5	0.2757
76.6	27	40	26	0.037	16	240	14.4	5	0.3000
76	22	41	28	0.037	16	240	14.4	5	0.1963
78	25.3	41	27	0.037	16	240	14.4	5	0.2511
77	26	41	27	0.037	16	240	14.4	5	0.2557
76	24	41	27.1	0.037	16	240	14.4	5	0.2194
75.1	23	41	27	0.037	16	240	14.4	5	0.2019
75.8	23	41	27	0.037	16	240	14.4	5	0.2060
75.6	21.9	41	26.2	0.037	16	240	14.4	5	0.1930
								average	0.2361
								std	0.0376

The table below shows the average and the standard deviation of each 10 repeats for each tested specimen:

Table 6- Average thermal conductivity for each specimen

	Dry	**Moist**
Average thermal conductivity – Dressing [w/mK]	0.1932 ± 0.0045	0.5630 ± 0.1043
Average thermal conductivity-Fabric [w/mK]	0.0644 ± 0.0170	0.2361 ± 0.0376

As seen in Table 6, the thermal conductivities of the dry dressing and the dry fabric are in the theoretical ranges, as it was suggested in the literature. Moreover, it can be seen that the thermal conductivities of the moist samples are larger than the thermal conductivities of the dry ones.

Two-tailed t-tests were performed on MATLAB for each tested material, to determine whether the thermal conductivities of the dry and moist samples are statistically similar.

The results of the t-test are shown in the following table:

Table 7- t-test results

Tested Material	**t-test result**	**P-value**
Dressing	1	1.5157e-09
Fabric	1	1.3208e-10

It can be seen from Table 7 that both t-tests rejected the null hypothesis that the thermal conductivities of the dry and moist samples are statistically similar, at a 5% significance level. It could be also seen by the small p-value of both tests.

4 Disscusion & Conclusion

The thermal conductivities of the dry dressing and the dry fabric which were calculated according to the data extracted in the experiment, appear to be in the theoretical ranges, as the literature suggests. For example, as for the thermal conductivity of the dry dressing, the calculated value is 0.1932 w/mK while the theoretical values suggested in the literature are 0.16-0.21 w/mK. When it comes to the dry fabric, the value which was calculated for the thermal conductivity, is 0.644 w/mM while the literature suggests 0.026-0.065 w/mK.

The hypothesis regarding the difference between the dry and the moist sample which has been tested is whether moist samples will have a larger thermal conductivity, compared to the dry ones. The results show that the thermal conductivities of the moist samples are indeed larger when compared to the thermal conductivity values of the dry ones. It has been found that the thermal conductivity rises as the moisture content does, primarily as a result of the sample's water particles transferring heat by conduction. [22]

By definition, the amount of heat transfer from the skin surface to the environment is determined by the dressing's thermal conductivity. Heat will build up at the skin-dressing interface of a dressing with low thermal conductivity. This will prevent heat from being released to the surface of the dressing and from there to the environment, such as through convection. [16]

It has been shown that the thermal conductivity values of various material types can be used to prevent pressure ulcers. Moreover, It has also been demonstrated that heat buildup at the skin-dressing interface may result in excessive skin heating, increasing the risk of developing pressure ulcers. [23] It can be said that the thermal conductivity of dressing materials is the most critical property regarding wound prevention and treatment, especially concerning preventative applications, where a dressing is used as an interface structure between the body and the support surface or between the body and a medical device. [16]

During the experiment, moist samples were prepared to simulate 100% of a moist environment. It has been demonstrated that moisture, using a suitable moist dressing speeds up wound healing by maintaining a consistent and moist environment- reducing dehydration, boosting angiogenesis and collagen formation, and accelerating the breakdown of dead tissue and fibrin. In addition to lessening discomfort, a moist environment enhances the appearance of the wound, which can be achieved by a moist wound dressing. [24,25] The rationale for this is that a humid atmosphere will assist the migration of the epithelium and speed up wound healing. [24] In a hot atmosphere, sweating evaporation rather than sweating is what cools the human body. Even though one sweats a lot in extremely hot and humid weather, he typically does not feel cool. This is derived from the fact that under high humidity conditions, sweat cannot evaporate and cannot remove latent heat from the skin; instead, it simply drips off the body. In dry environments, sweat is often dissipated by the heat provided by the skin surface (insensible evaporation), without soaking the skin surface. As a result of the

cooling action, the body temperature sharply increases. [26] Wet clothing offers far less insulation than dry clothes between the body and the environment, which helps to prevent convective and conductive heat loss. [27] The risk of getting hypothermia increases with long-term exposure to cold, wet, or windy conditions or some of these conditions combined. These are most common during endurance competitions, outdoor team sports (including soccer and football), mountaineering, hiking, and military maneuvers, as well as in jobs that require spending extended amounts of time outside or in cold environments. [28]

Several factors could have affected the results and led to large standard deviations and other errors while experimenting. First of all, the heat-flow meter has a large device error since it can only show the first digit to the right of the decimal point of the value measured which causes rounding the numbers indicating the temperature. For instance, it can be shown that the majority of the measured values for room temperature were comparable. However, throughout the experiment, the window was open and the air conditioner was turned on, both causing temperature variations that, after the software rounded the temperature numbers, revealed similar values for the temperature even though they were not the same. The heat-flow meter device is moreover extremely sensitive to small motions in its surrounding. For instance, even when the machine had previously established a quasi-steady state with consistent values, the results are impacted when one gets close to the table on which the device is positioned. Additionally, even when the sample was placed on the apparatus without moving, the readings for the weight of the moist samples were unstable and fluctuated. This might have had an impact on the sample's moisture content and the amount of time needed to prepare each moist sample until it could have been placed on the heat-flow meter.

A variety of suggestions can be made that might improve the experiment. A sample of an IDF coat could be utilized as well. It is made of fleece, specifically two layers of high-density, lightweight microfleece fabric. [18] Moreover, the influence of the Young modulus for each sample could be examined. The idea of local padding applied to a vulnerable anatomical site directly addresses the underlying biomechanical factors ultimately leading to pressure ulcers, that is, by reducing the stiffness gradient between the support surface or device materials and the skin and increasing the contact area for transferring mechanical loads, tissue stress concentrations are redistributed and reduced in intensity. [16] Another thing that can be changed is the area

of the tested specimen in contact with the superior/inferior blocks (according to formula (1)).

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